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1 MAPRO Full Analysis Test Run

Command executed:

```
bash ../scripts/driver.sh \  
  -n ../data/neidb \  
  < ../data/test_list.txt
```

The driver script processes each organism listed in `test_list.txt` independently.

1.1 1. Aquifex aeolicus VF5 (aae)

Output:

```
Analyzing aae Aquifex aeolicus VF5  
Collecting 1 genome record  
Downloading: tdata.zip  
...  
Error: There are no genome assemblies that match your query.
```

Explanation:

- The type strain was found in the Neighbors database.
- Only one target genome accession was available.
- NCBI Datasets attempted to download the genome.
- No complete/chromosome-level assembly was available.
- As a result, no genome FASTA file was downloaded.

Consequences:

```
ls: cannot access 'aae/targets/'  
ls: cannot access 'aae/neighbors'
```

Since no genomes were available, MAPRO could not build a phylogeny and therefore skipped marker discovery for this species.

Result:

- Analysis failed.
- No markers produced.
- No primers produced.

—

1.2 2. *Agrobacterium fabrum* C58 (afa)

Output:

```
Collecting 20 genome records  
Collecting 68 genome records
```

Explanation:

- 20 genomes were identified as target genomes.
- 68 genomes were identified as neighboring genomes.

The genomes were downloaded and rehydrated successfully.

Output:

```
Found 20 of 20 files for rehydration  
Completed 20 of 20
```

```
Found 68 of 68 files for rehydration  
Completed 68 of 68
```

A phylogeny was constructed and a representative target genome was selected:

using tGCF_003667945.1_ASM366794v1_genomic.fna as target representative

Marker discovery (Fur):

Step	Sequences	Length	Ns
Subtraction ₁	5305	1487141	0
Intersection	2188	469635	18938
Subtraction ₂	55	39423	26398

Interpretation:

- Subtraction₁: sequences present in target genomes but absent from neighbors.
- Intersection: sequences shared among all target genomes.
- Subtraction₂: final candidate marker regions.

Final result:

- 55 marker candidates.
- Total marker length = 39,423 bp.

—

1.3 3. *Aliivibrio fischeri* ES114 (afi)

Output:

Collecting 12 genome records
Collecting 15 genome records

Meaning:

- 12 target genomes.
- 15 neighboring genomes.

Representative genome:

using tGCF_000241785.1_ASM24178v1_genomic.fna as target representative

Marker discovery:

Subtraction_1	4484	3626854	1202
Intersection	6041	2001815	169339
Subtraction_2	363	98908	28173

Interpretation:

- Large amount of target-specific sequence was found.
- After filtering and intersection steps, 363 candidate markers remained.

Final result:

- 363 marker candidates.
- Total marker length = 98,908 bp.

—

1.4 4. *Aeromonas hydrophila* ATCC 7966 (ahy)

Output:

Collecting 4 genome records
Collecting 5 genome records

Meaning:

- 4 target genomes.
- 5 neighboring genomes.

Representative genome:

using tGCF_000014805.1_ASM1480v1_genomic.fna as target representative

Marker discovery:

Subtraction_1	1597	530113	0
Intersection	330	59775	4225
Subtraction_2	20	11874	776

Interpretation:

- Much fewer marker regions survived filtering.

- Only 20 candidate markers remained.

Final result:

- 20 marker candidates.
- Total marker length = 11,874 bp.

—

1.5 5. *Amycolatopsis mediterranei* U32 (ame)

Output:

Collecting 5 genome records

Collecting 49 genome records

Meaning:

- 5 target genomes.
- 49 neighboring genomes.

Representative genome:

using tGCF_000196835.1_ASM19683v1_genomic.fna as target representative

Marker discovery:

Subtraction_1	12457	4250144	0
Intersection	11520	4038681	200
Subtraction_2	539	1228442	643033

Interpretation:

- Very large amount of sequence survived the first filtering step.
- More than 1.2 Mb of candidate marker sequence remained.
- This species appears to have many regions distinguishable from its neighbors.

Final result:

- 539 marker candidates.
- Total marker length = 1,228,442 bp.

—

1.6 General Workflow

For each organism, MAPRO performs:

1. Identify target and neighboring taxa using the Neighbors database.
2. Download genomes from NCBI.
3. Build a phylogeny.
4. Select a representative target genome.
5. Create a Fur database.
6. Run Fur marker discovery.
7. Produce marker sequences.
8. Design PCR primers from the discovered markers.

The first organism (*Aquifex aeolicus*) failed because no suitable assembly was available from NCBI.

All subsequent organisms completed successfully and entered the marker-discovery stage.